Kalman Filter Theory and Applications Equation Drilldown

Michael L. Carroll

March 31, 2021

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Part 1
Equation Drilldown
Topics

- Mathematical Formulation of the Problem

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- Drilldown on State Dynamics and Covariance Extrapolation Equations

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- The Python and MATLAB files

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- Scalar Example: Resistor Revisited

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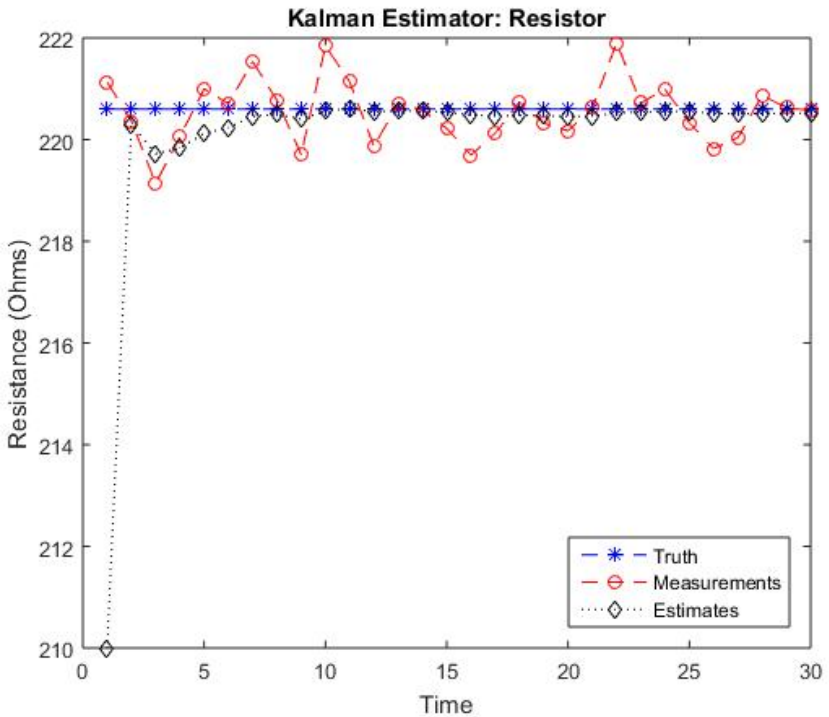
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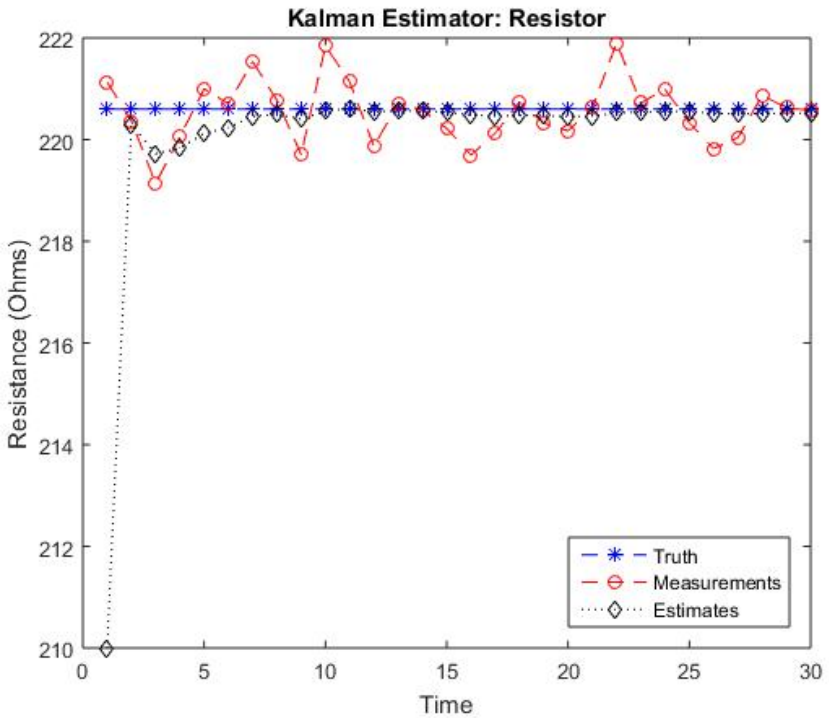
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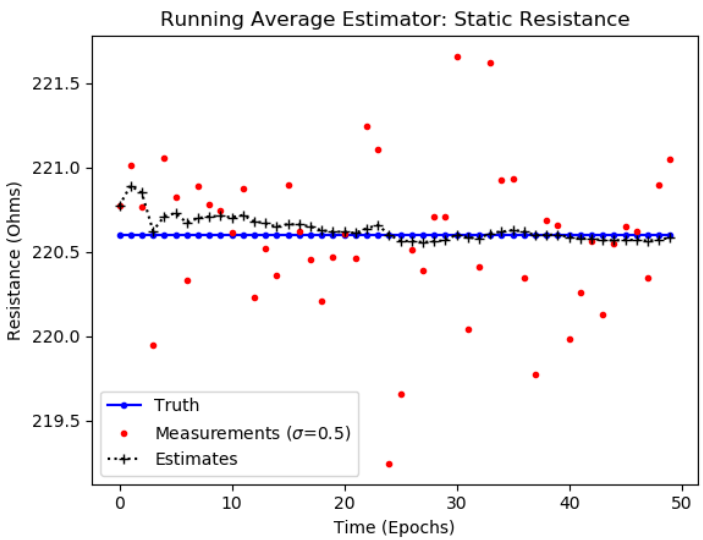


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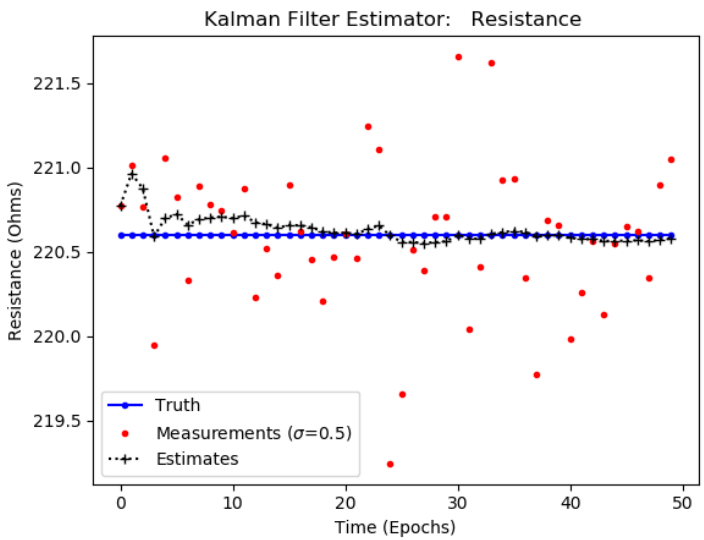


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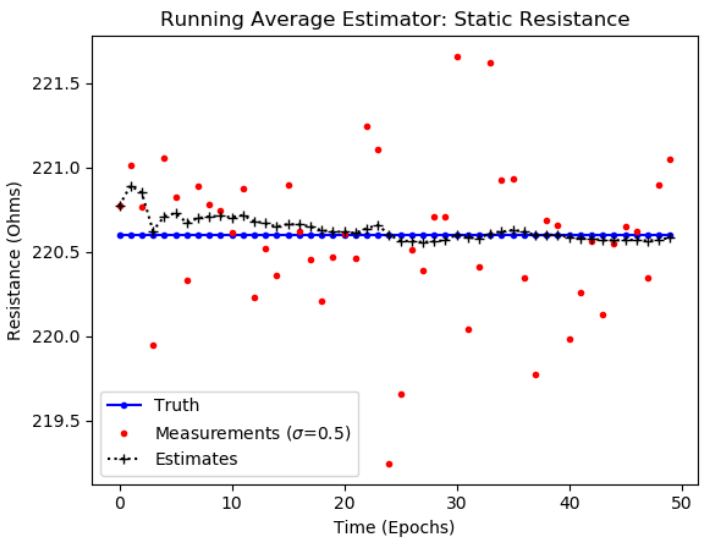
(a) Simple Running Average Estimator



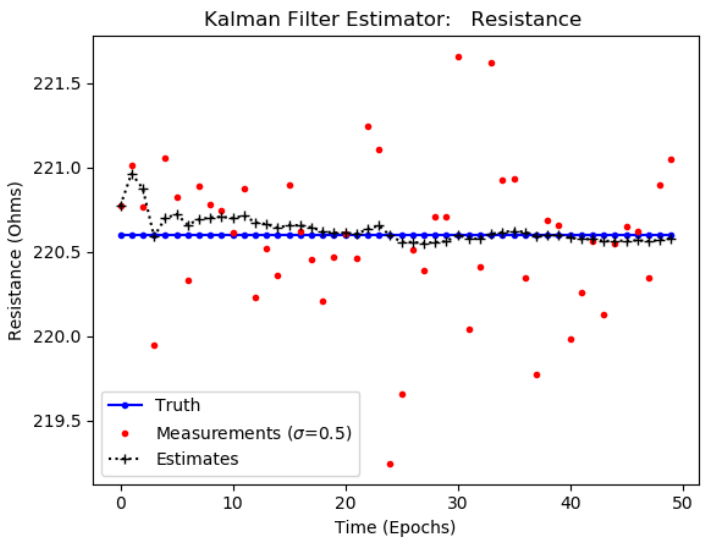
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Figure: Comparing Running Average and Kalman Estimators

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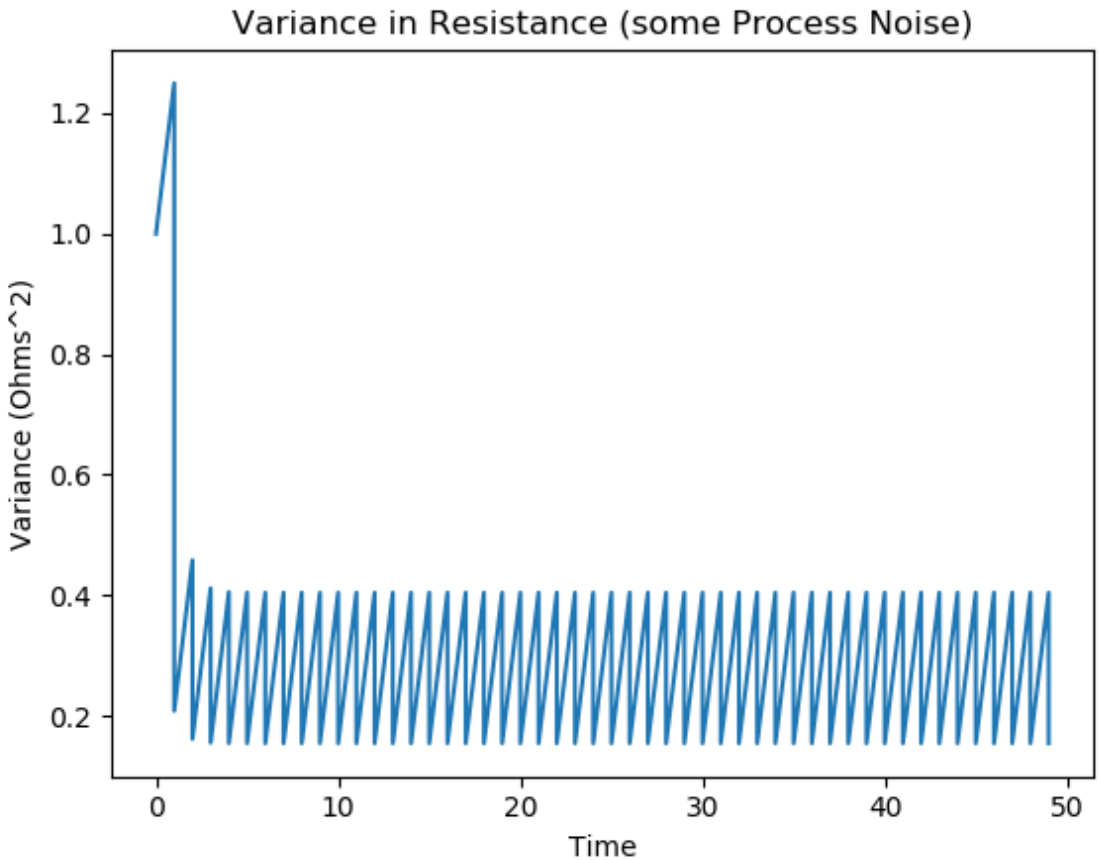
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Kalman Sawtooth Plots

By interleaving the pre- and post-update elements of P over a run (typically only the diagonal elements), we can obtain Kalman sawtooth plots.

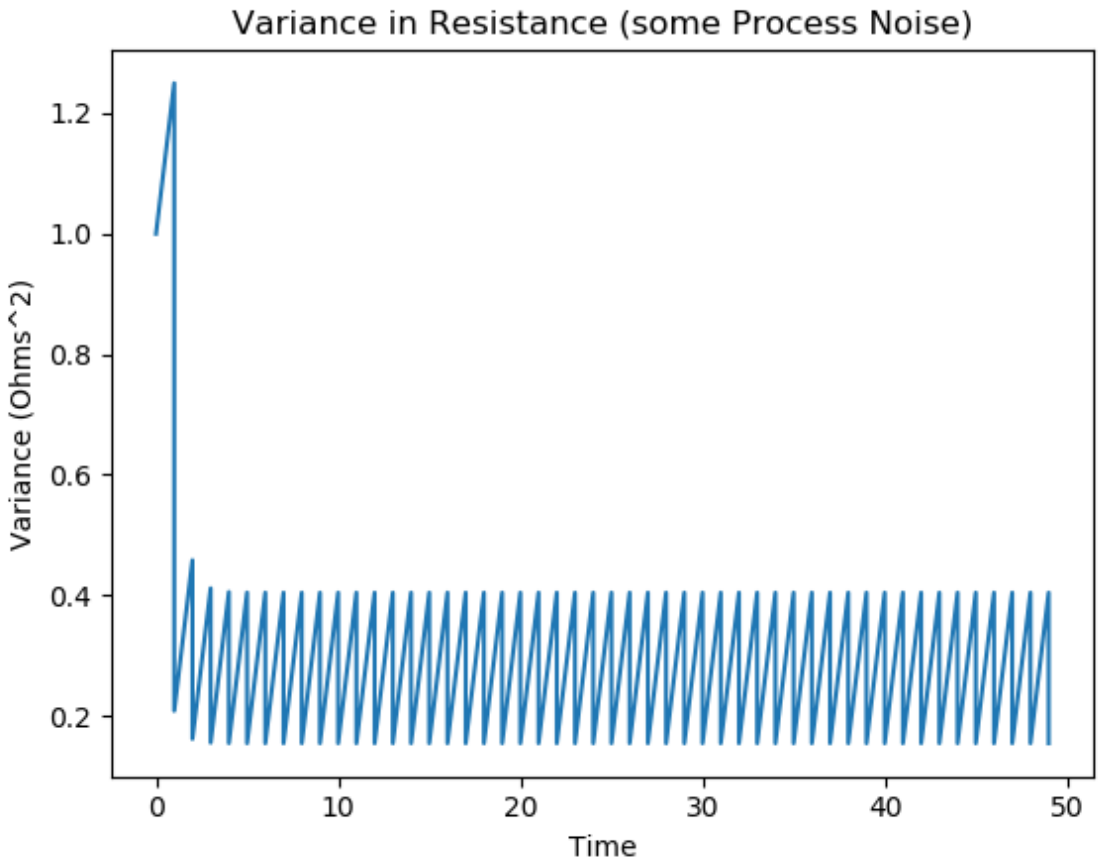
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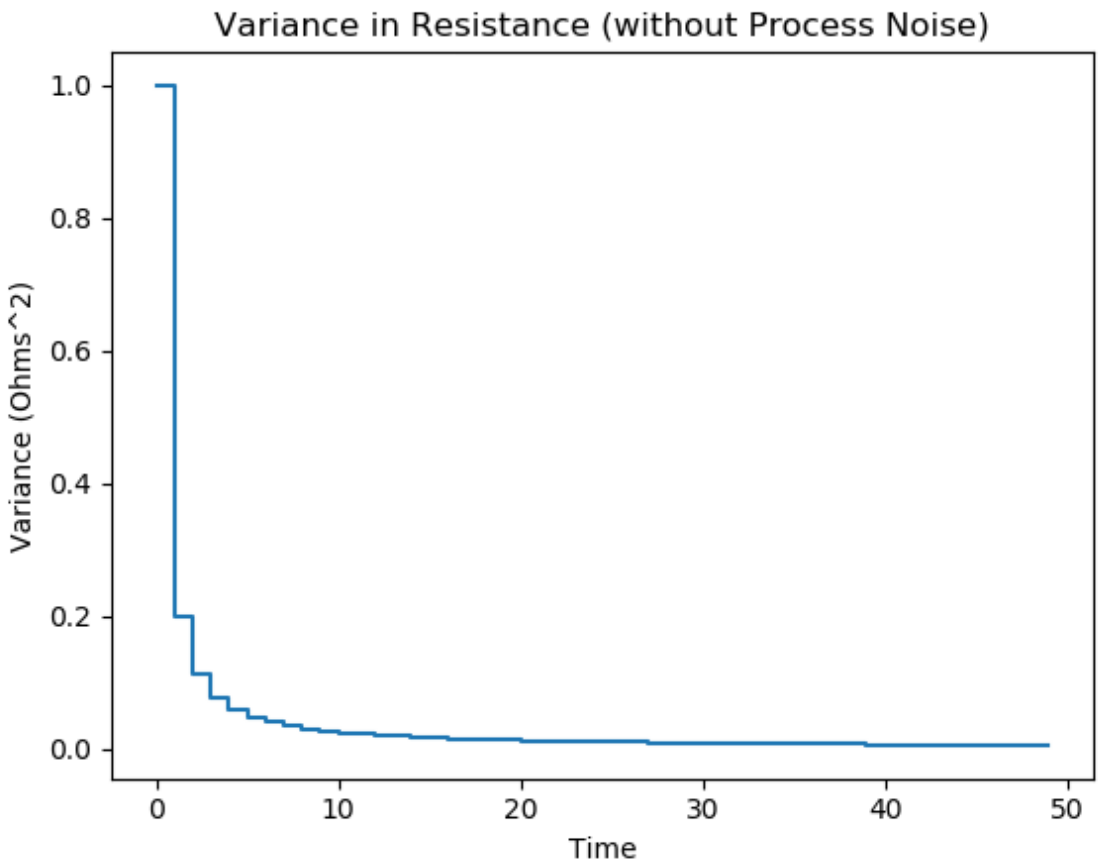


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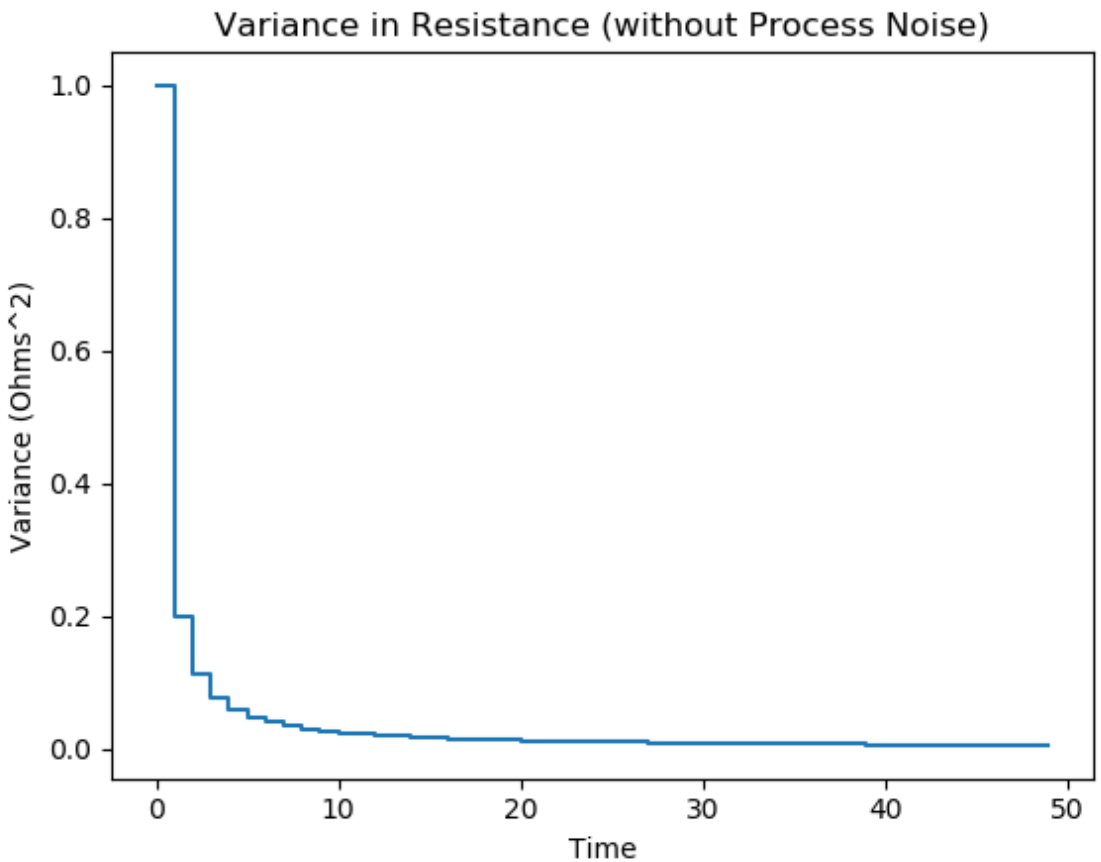
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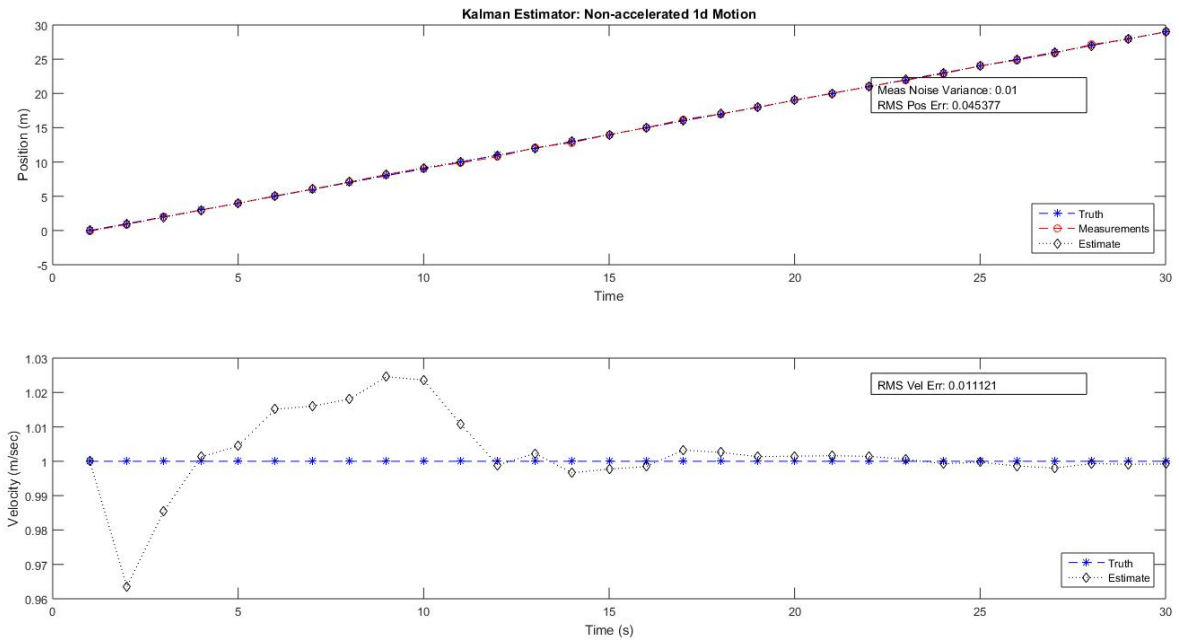
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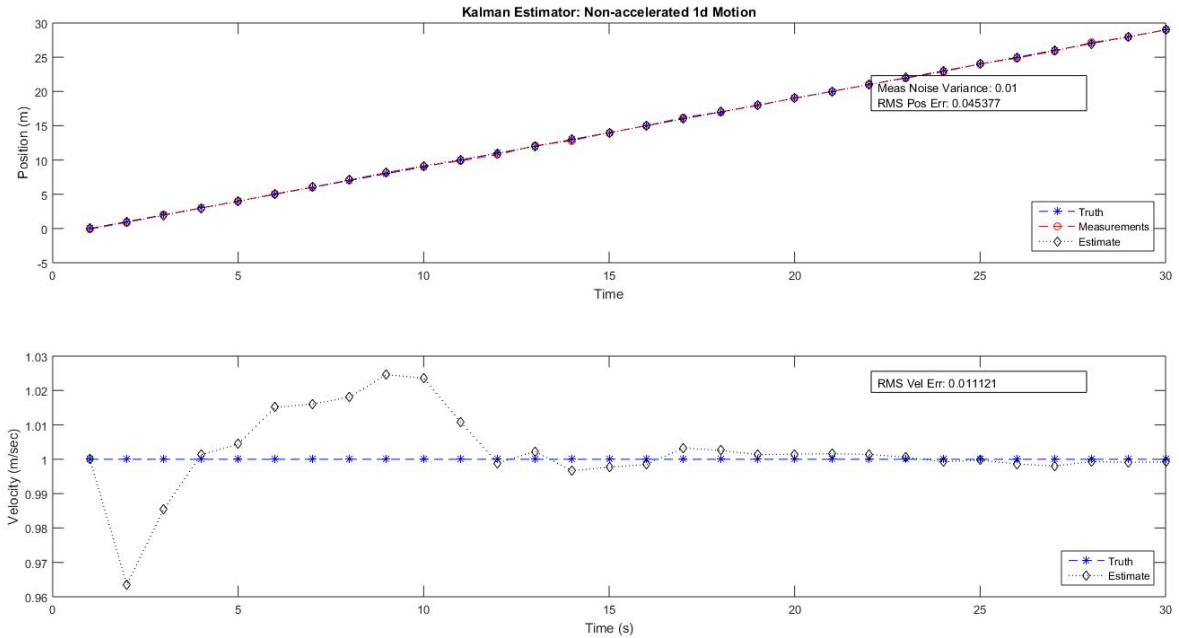
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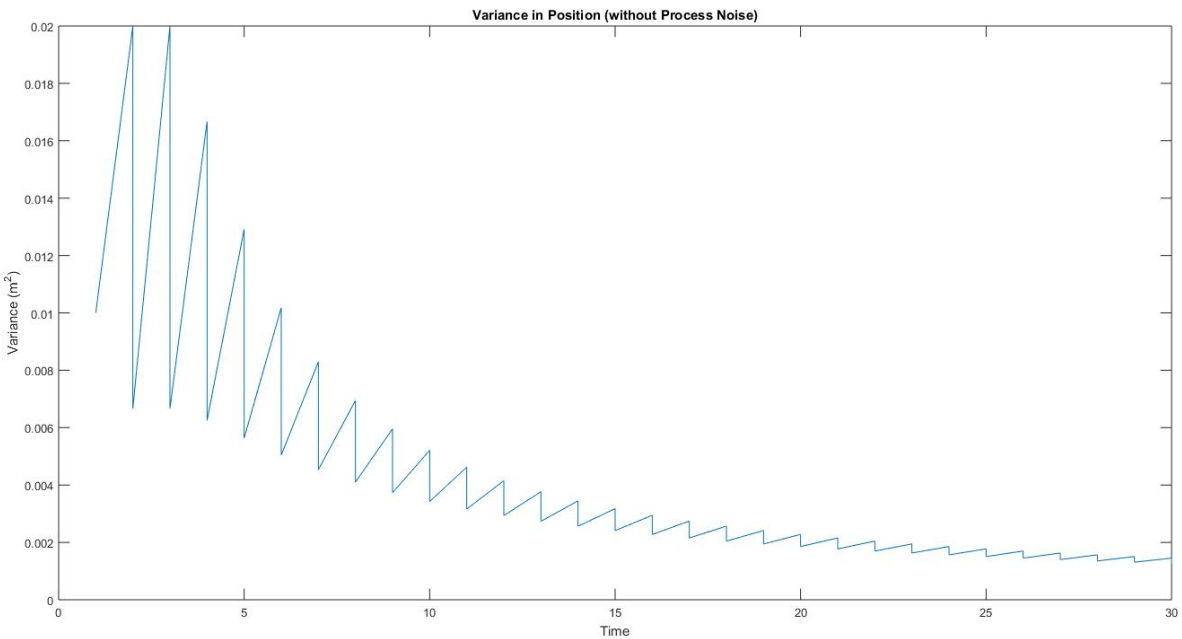
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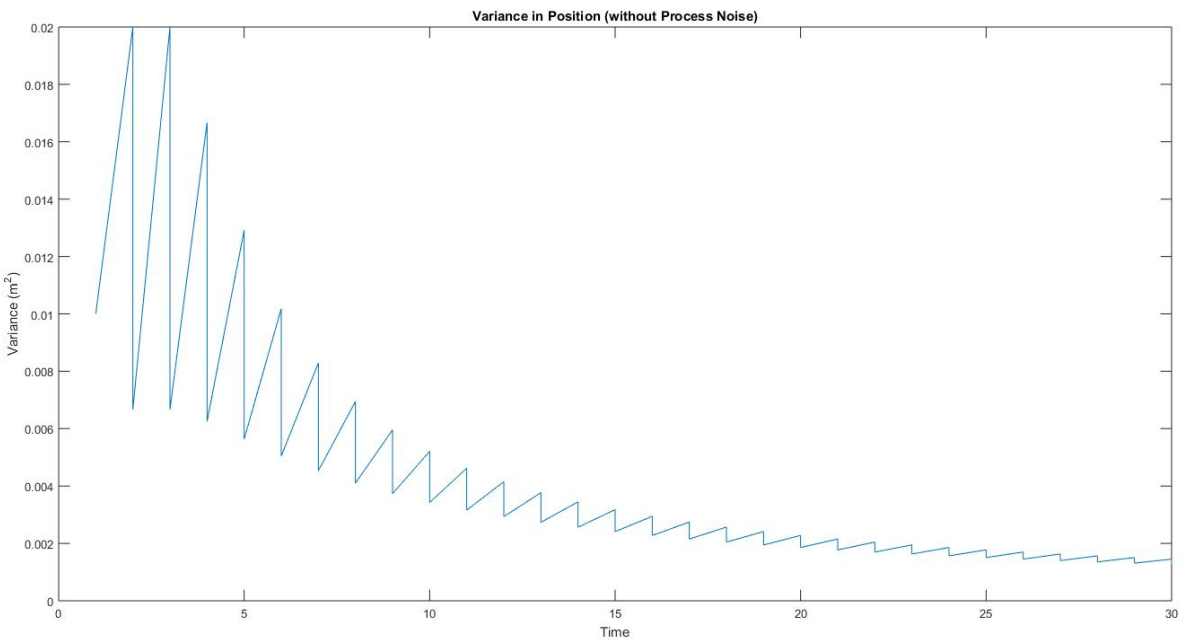
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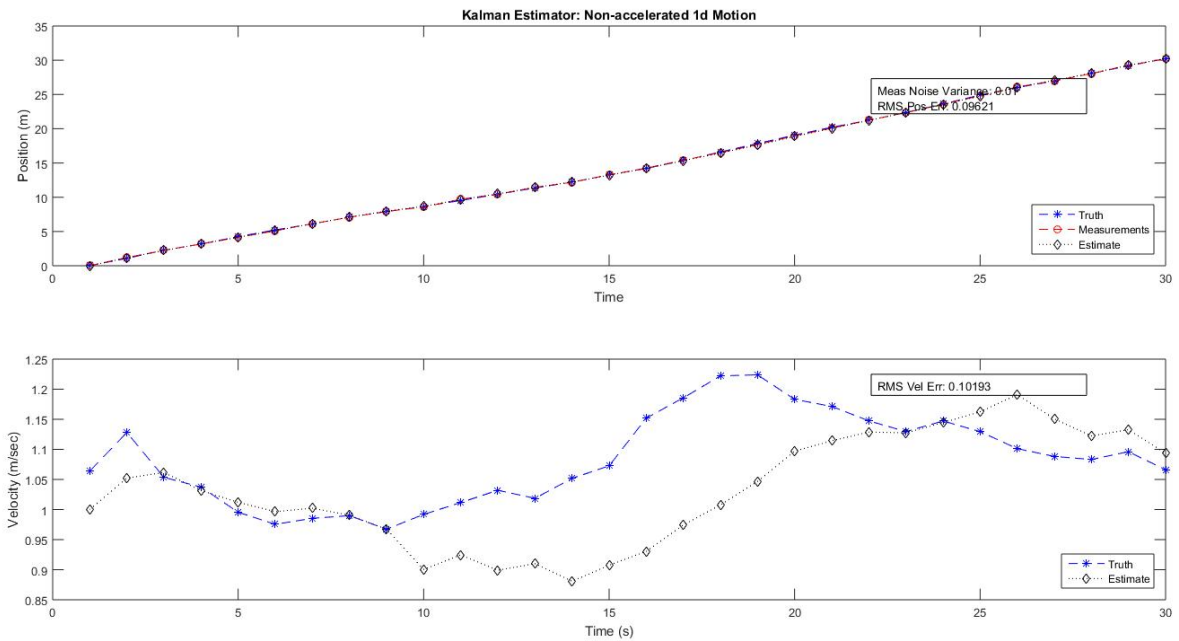
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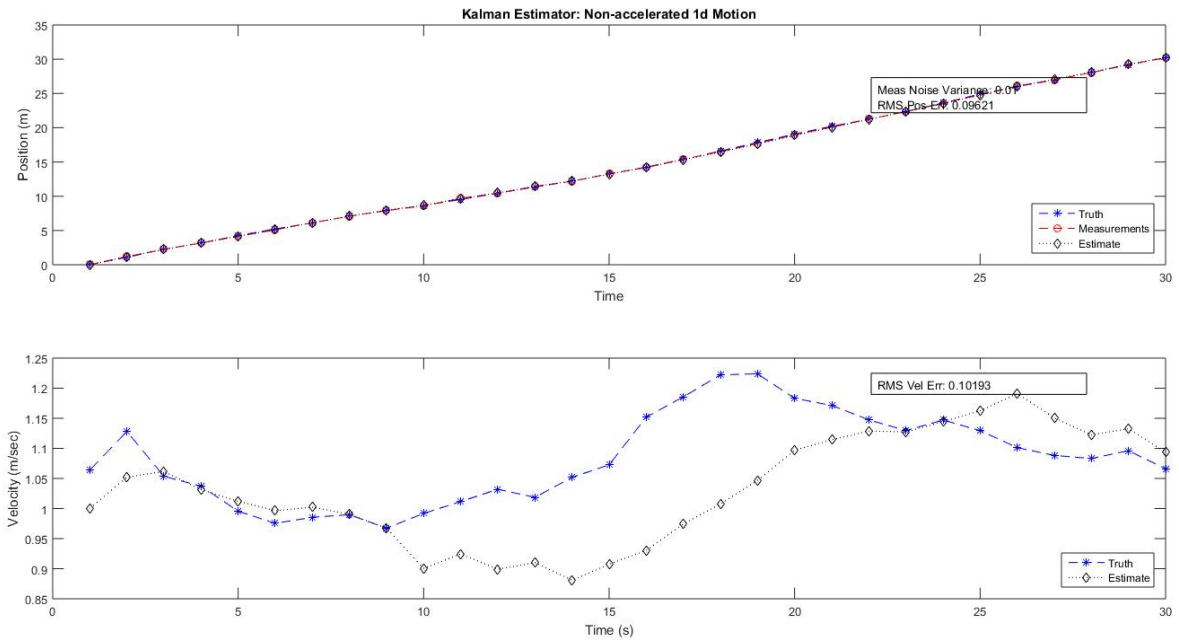


Non-accelerated Motion (constant velocity): Filter Performance (with Process Noise)

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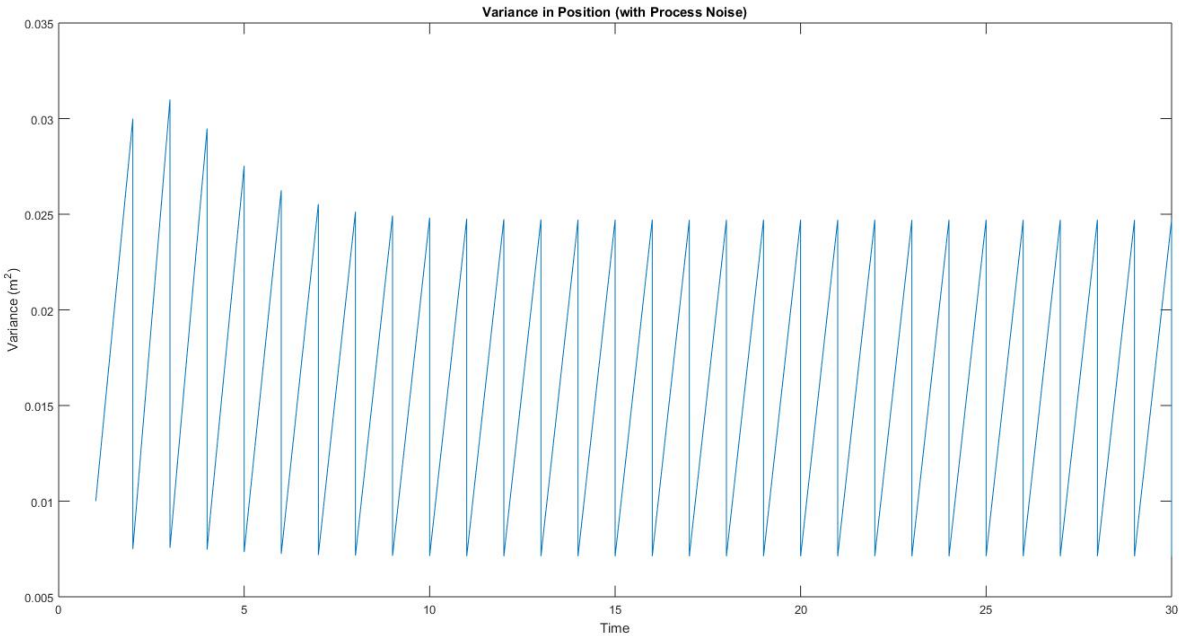


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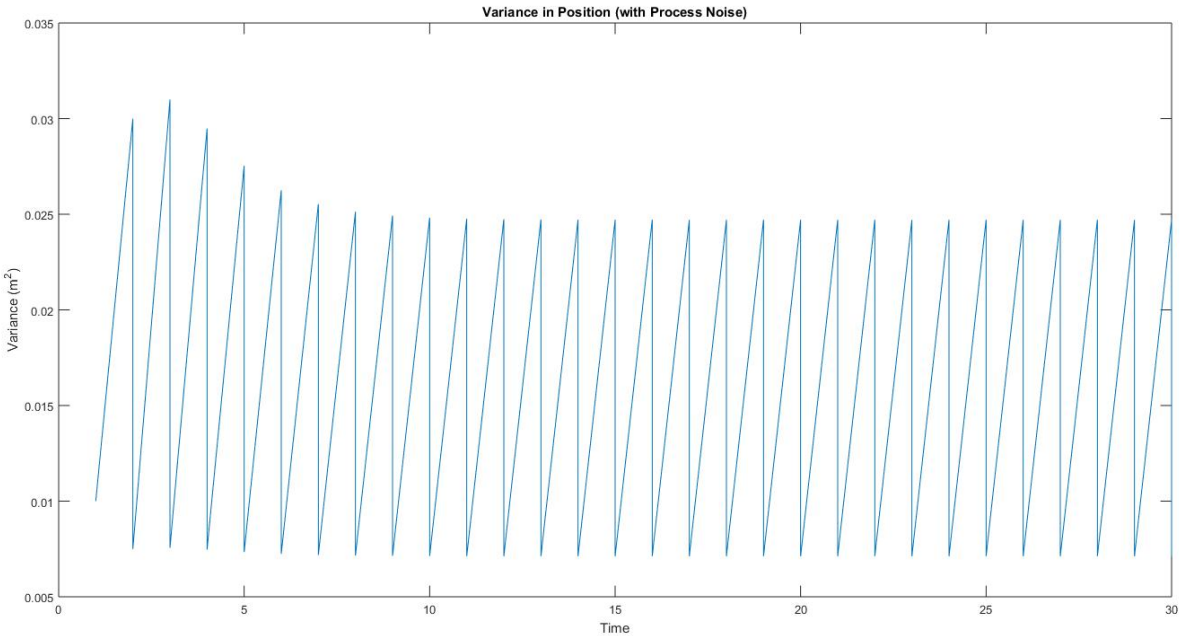


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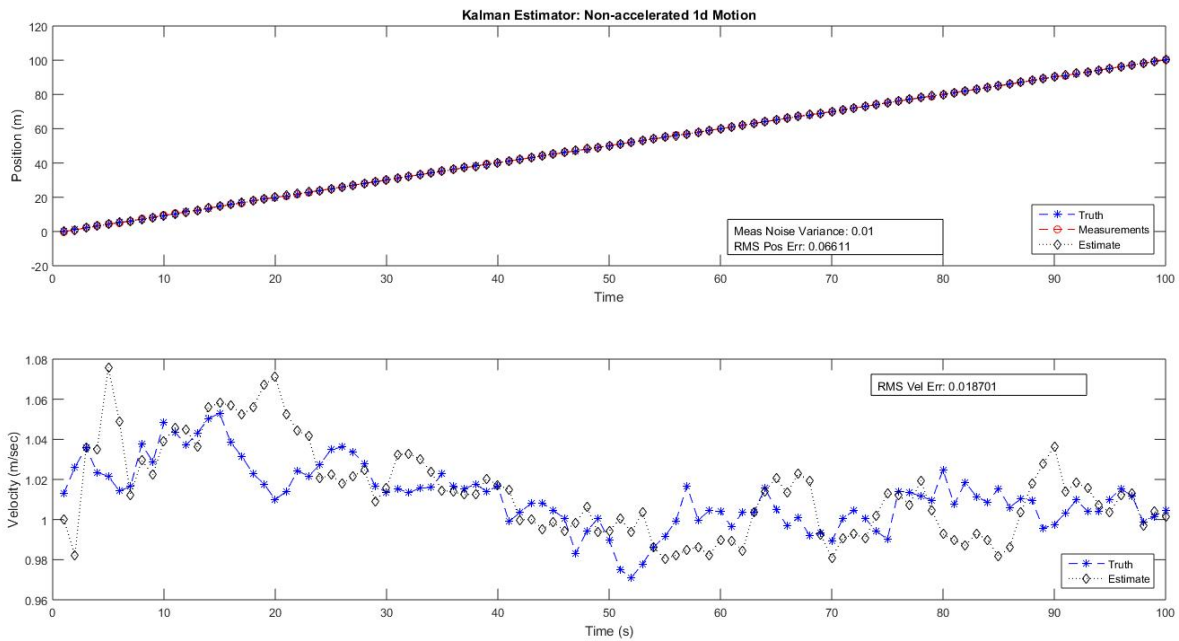


Non-accelerated Motion (constant velocity): Filter Performance (100 samples; less Process Noise)

Position Process Noise: 0.001; Velocity Process Noise: 0.0001

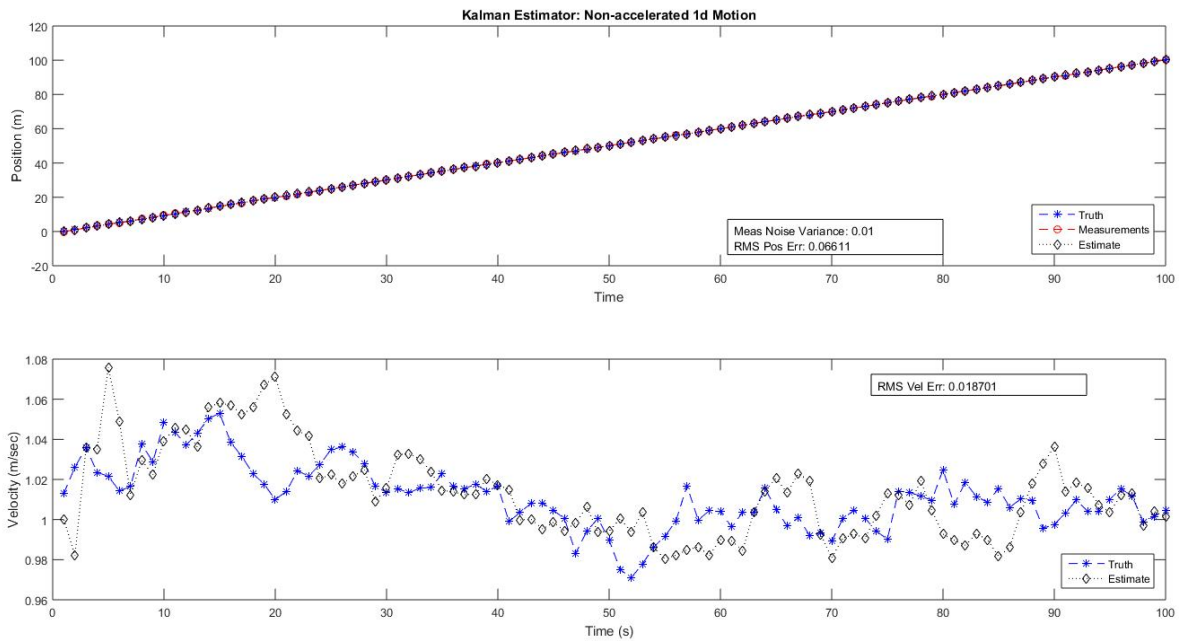
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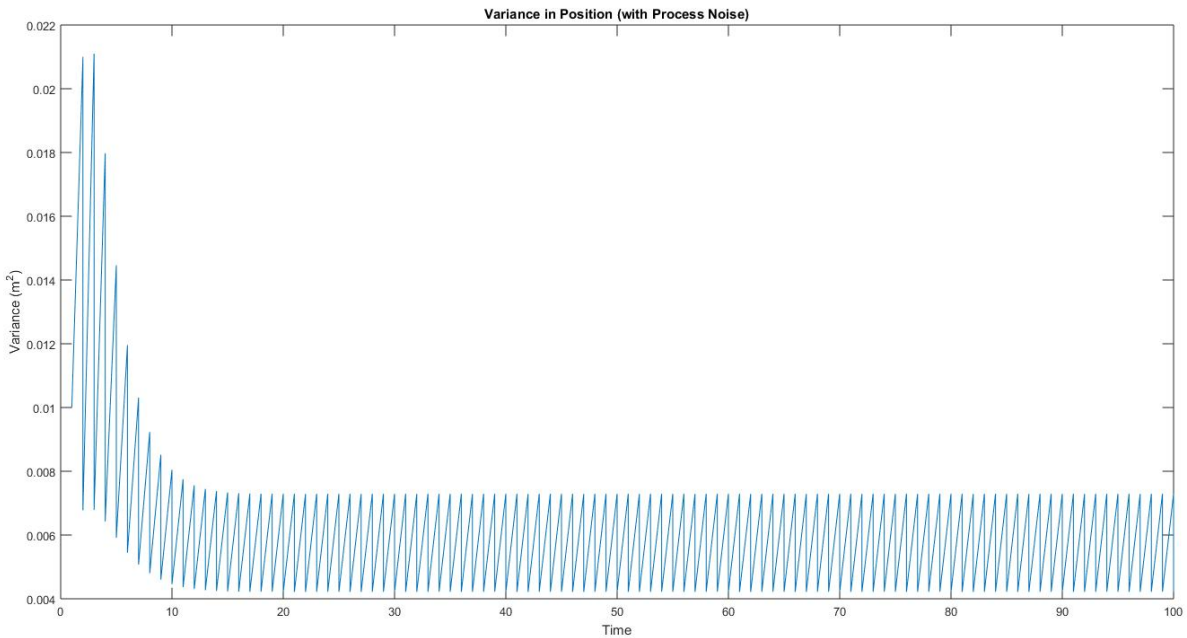


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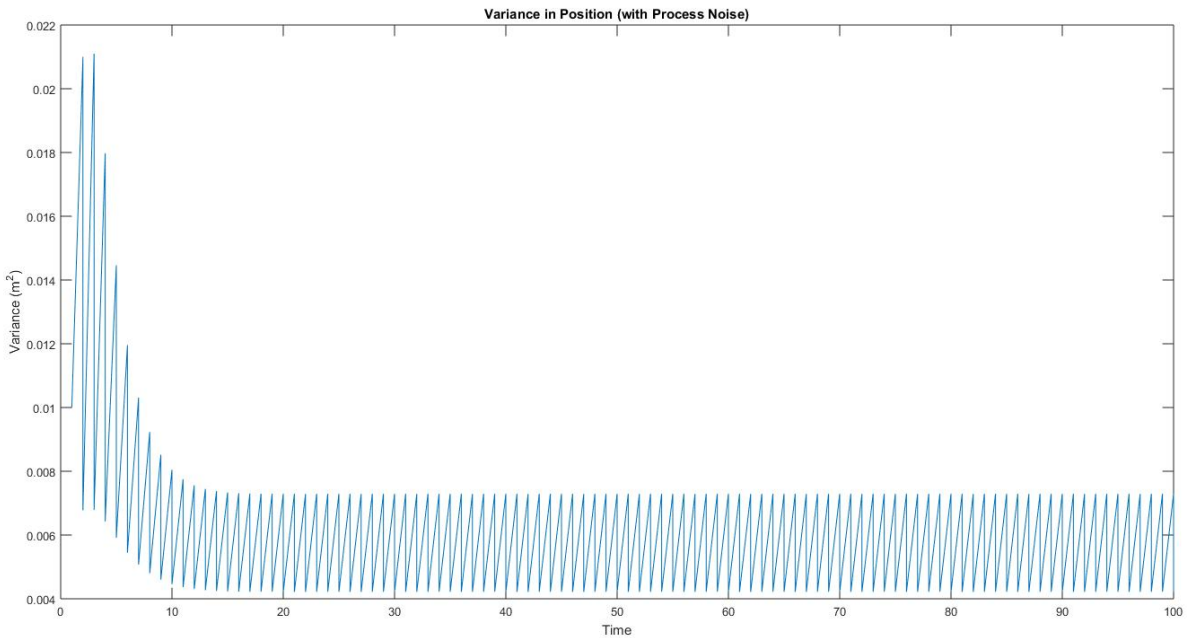
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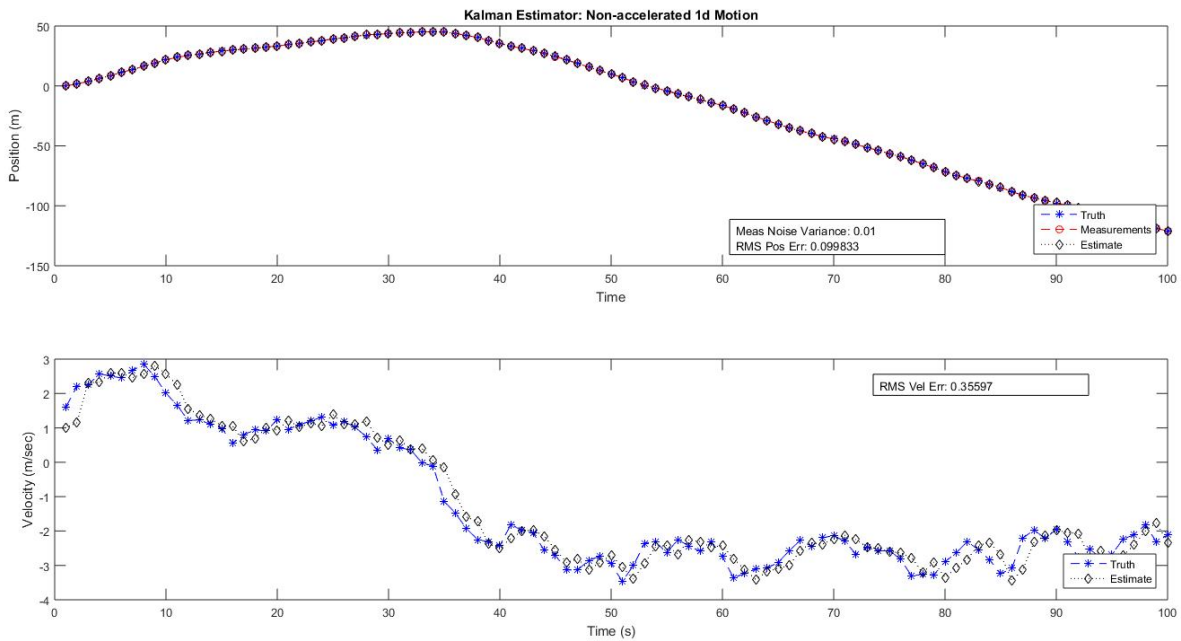


Non-accelerated Motion (constant velocity): Filter Performance (100 samples; large Process Noise)

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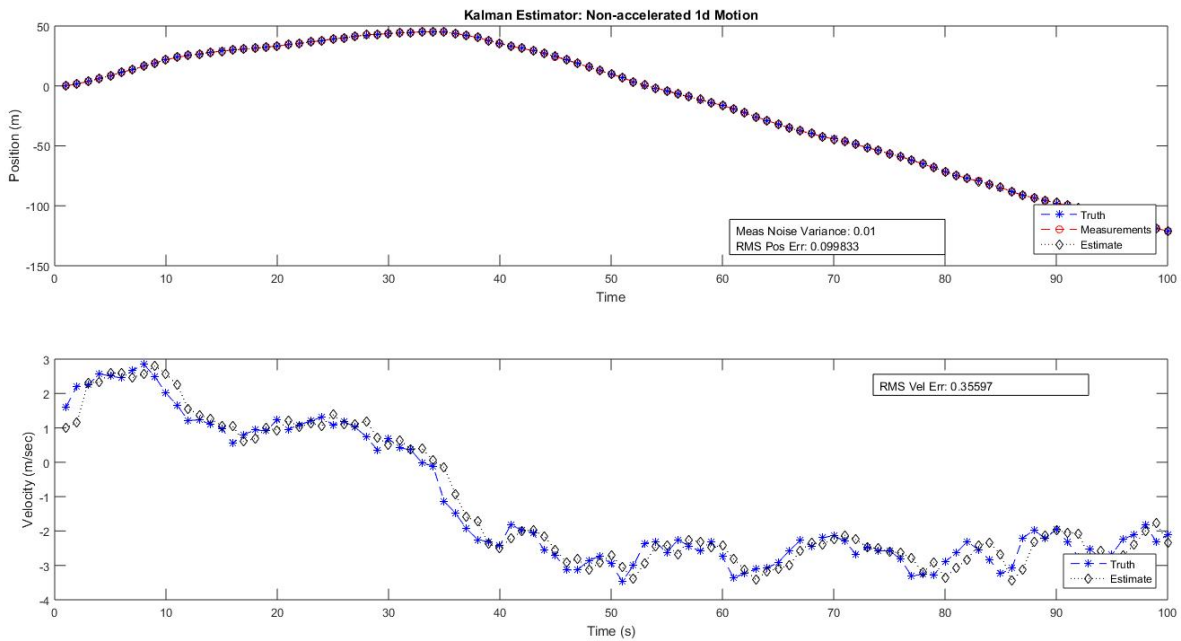
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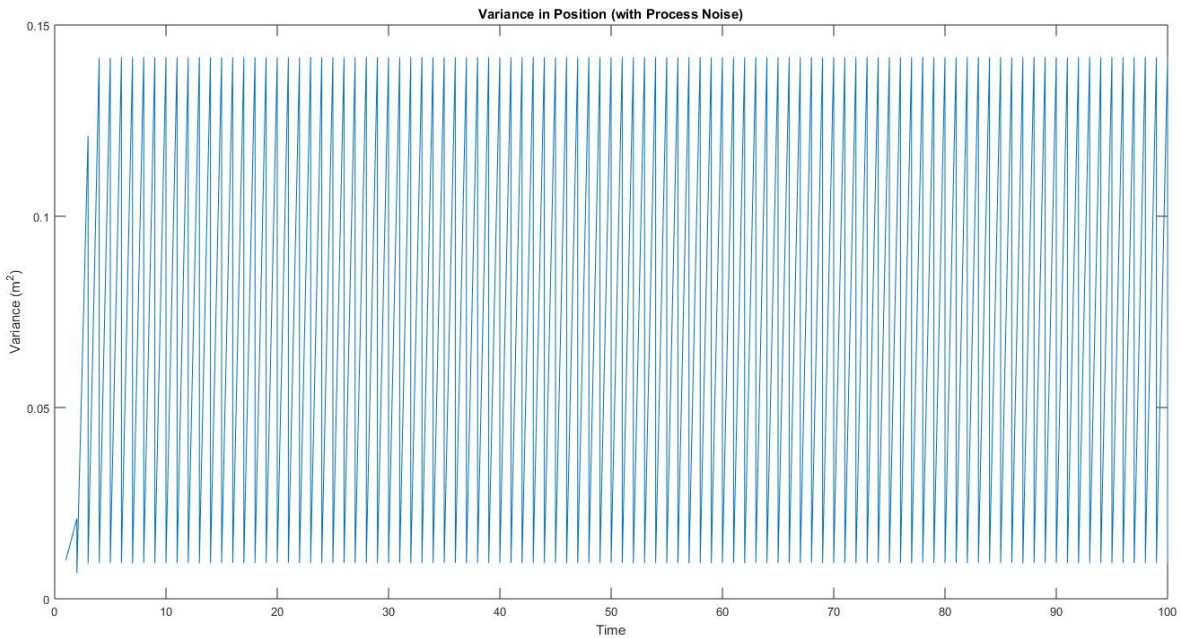


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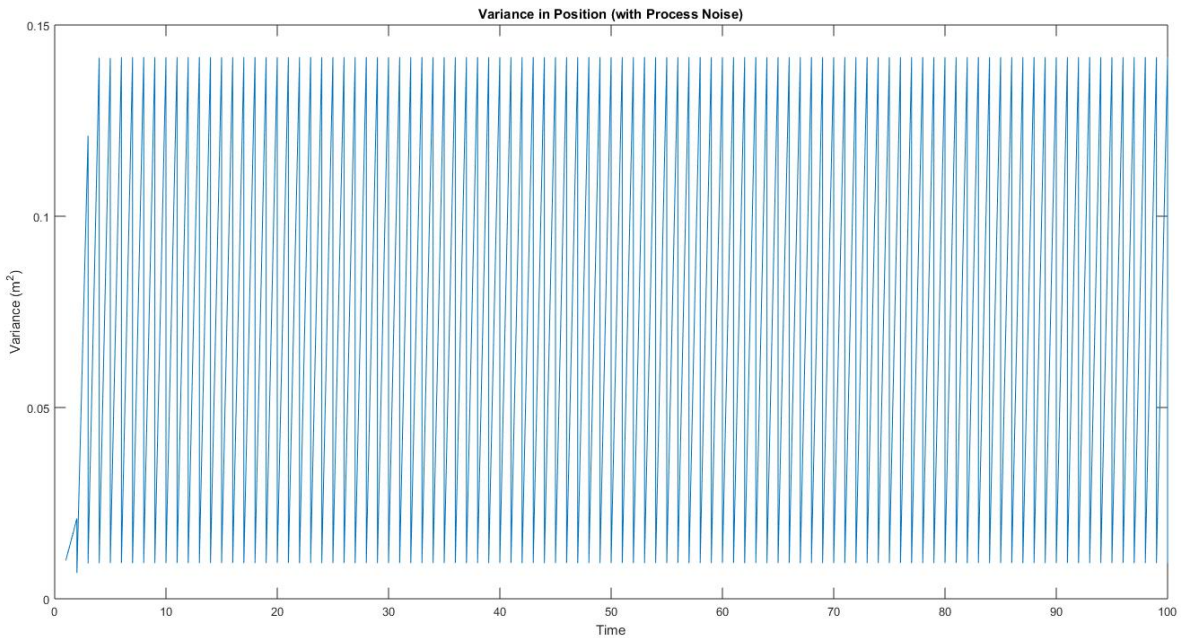
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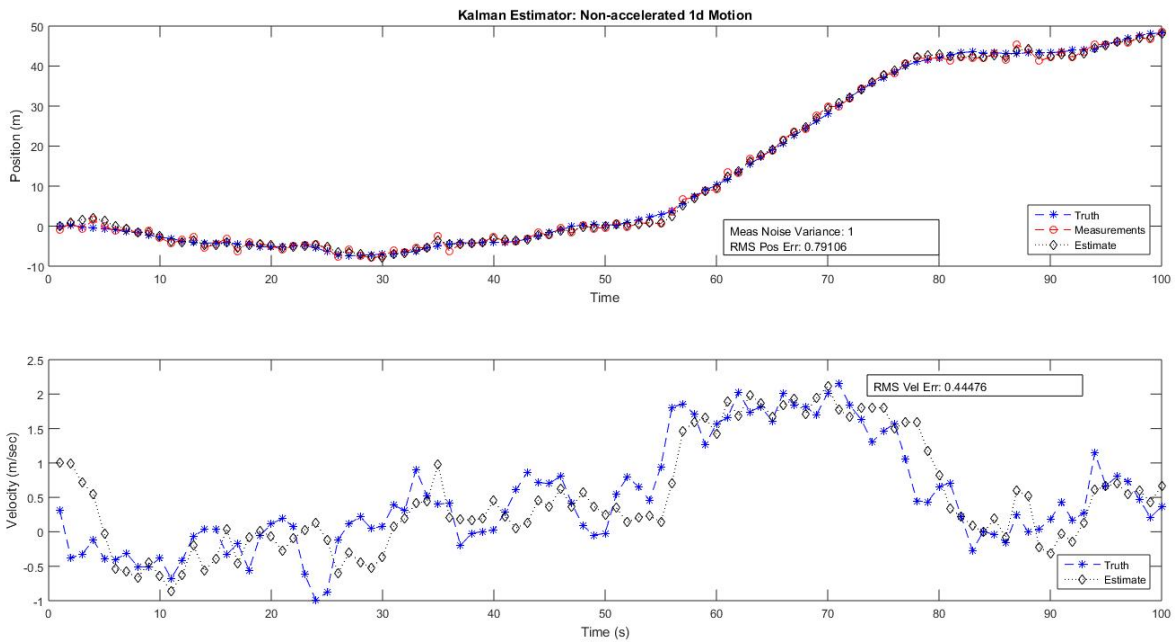


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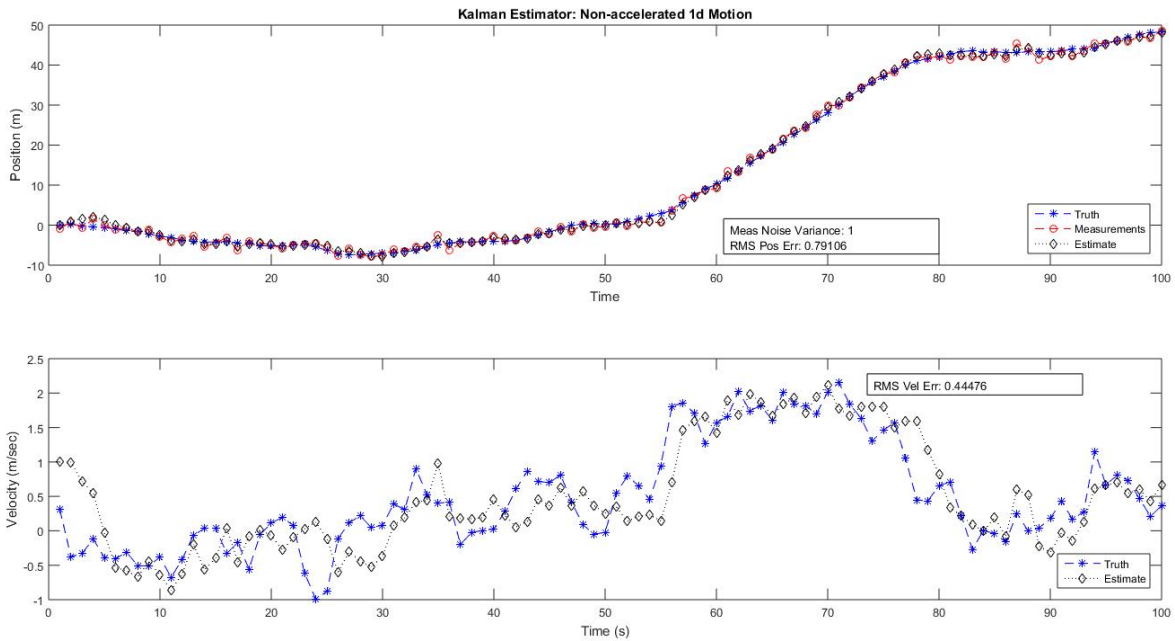
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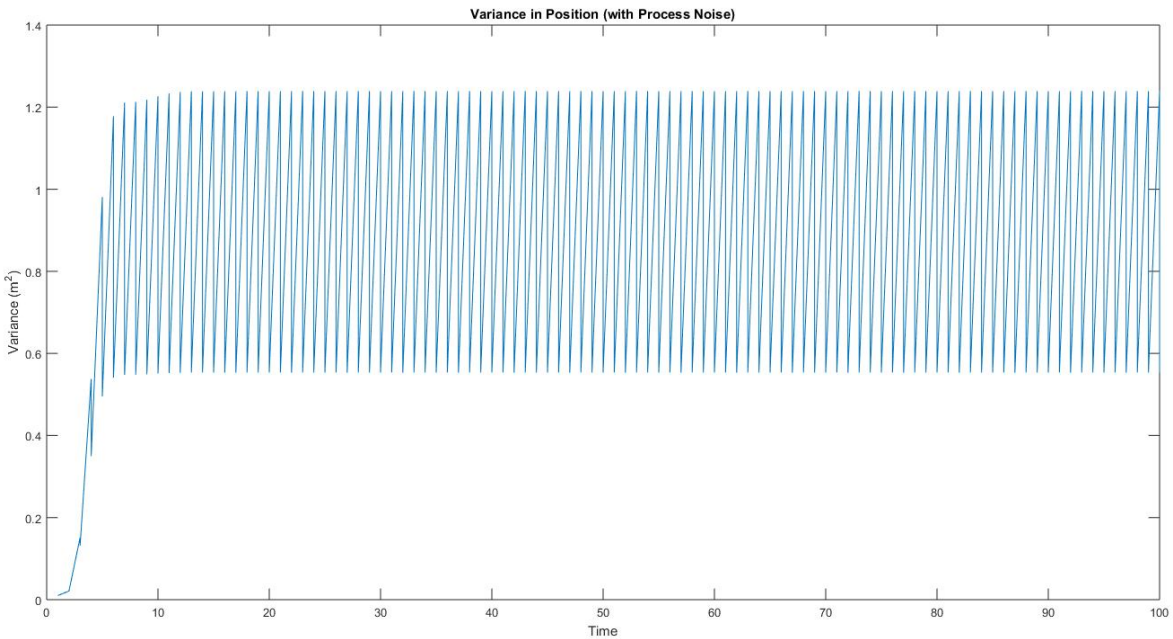


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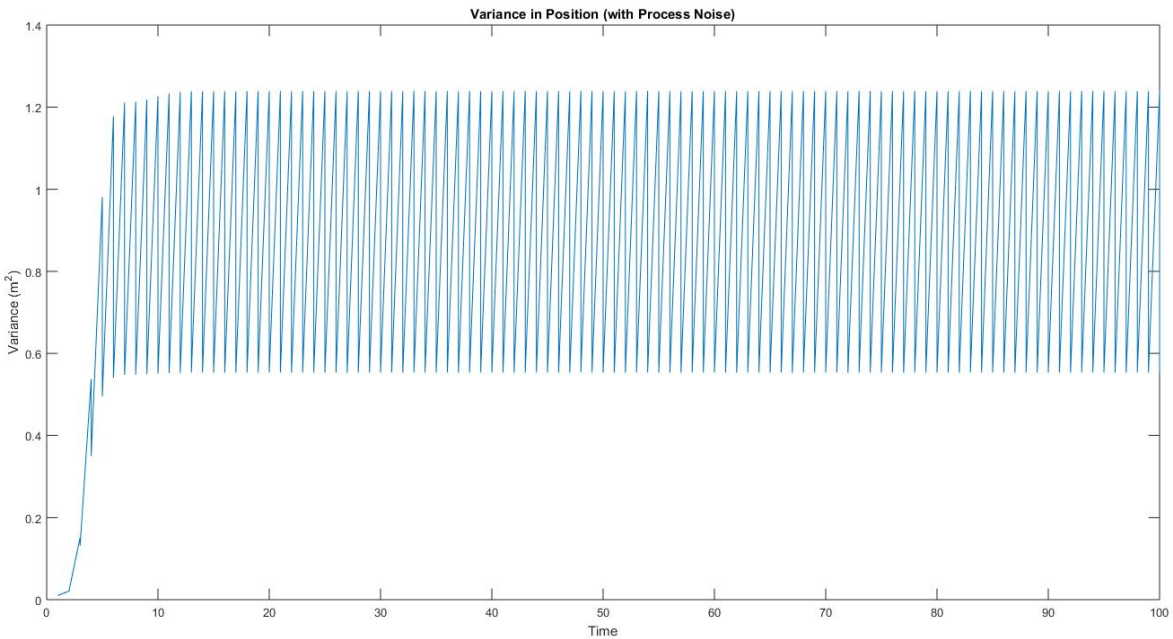
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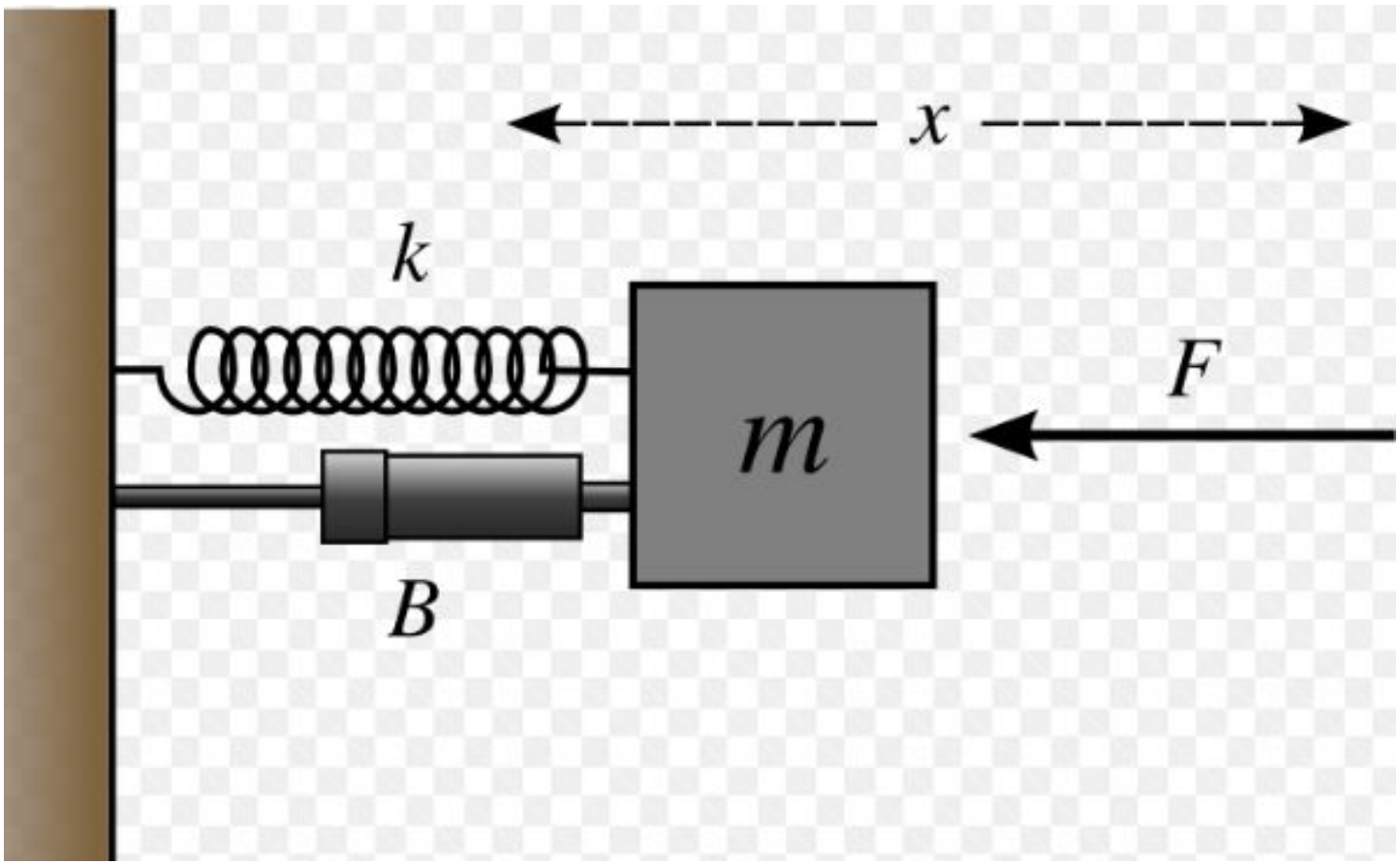
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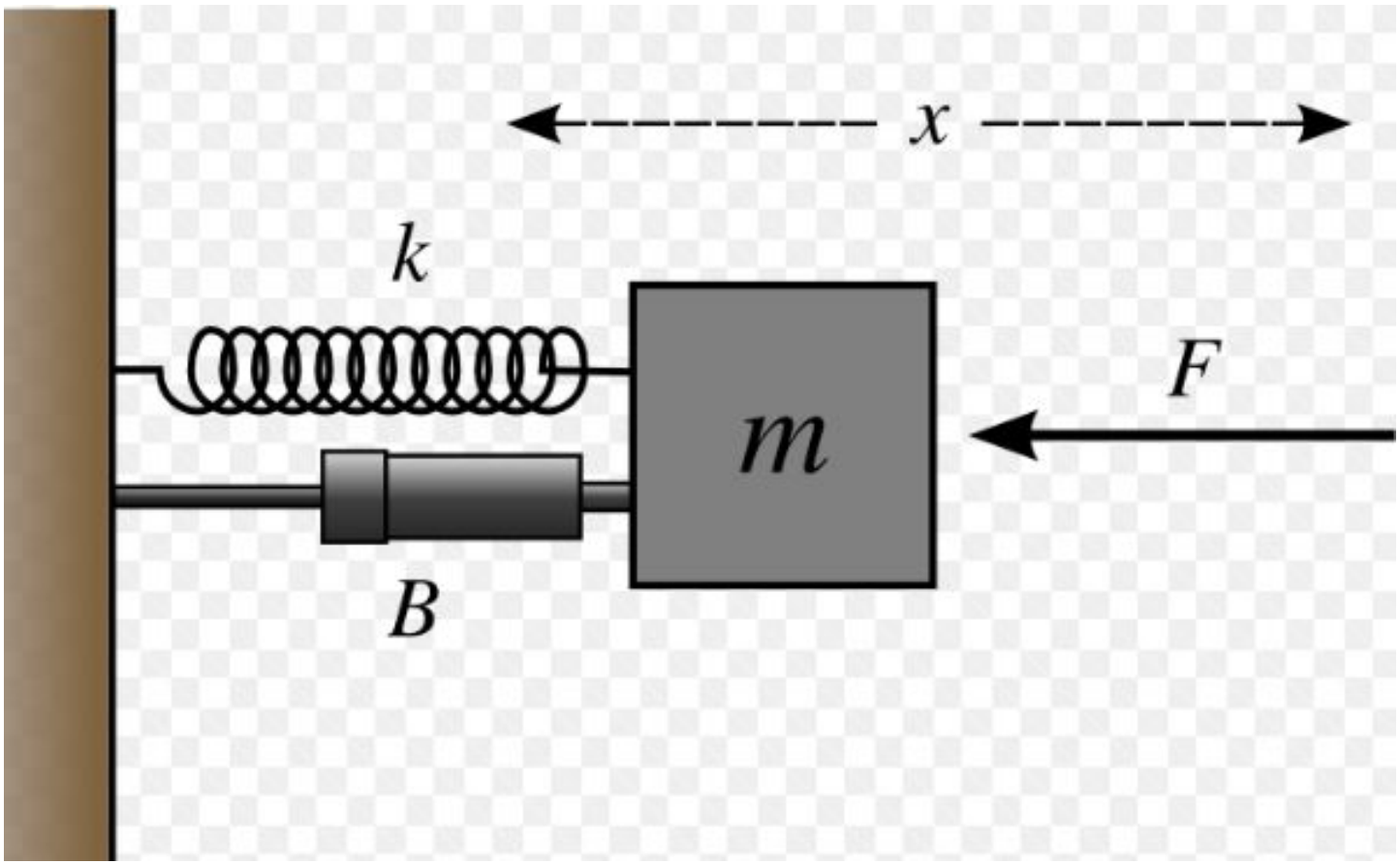


Damped Harmonic Oscillator: Spring, Mass, Damper System

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where we have once again suppresed the +/- superscripts and k's.

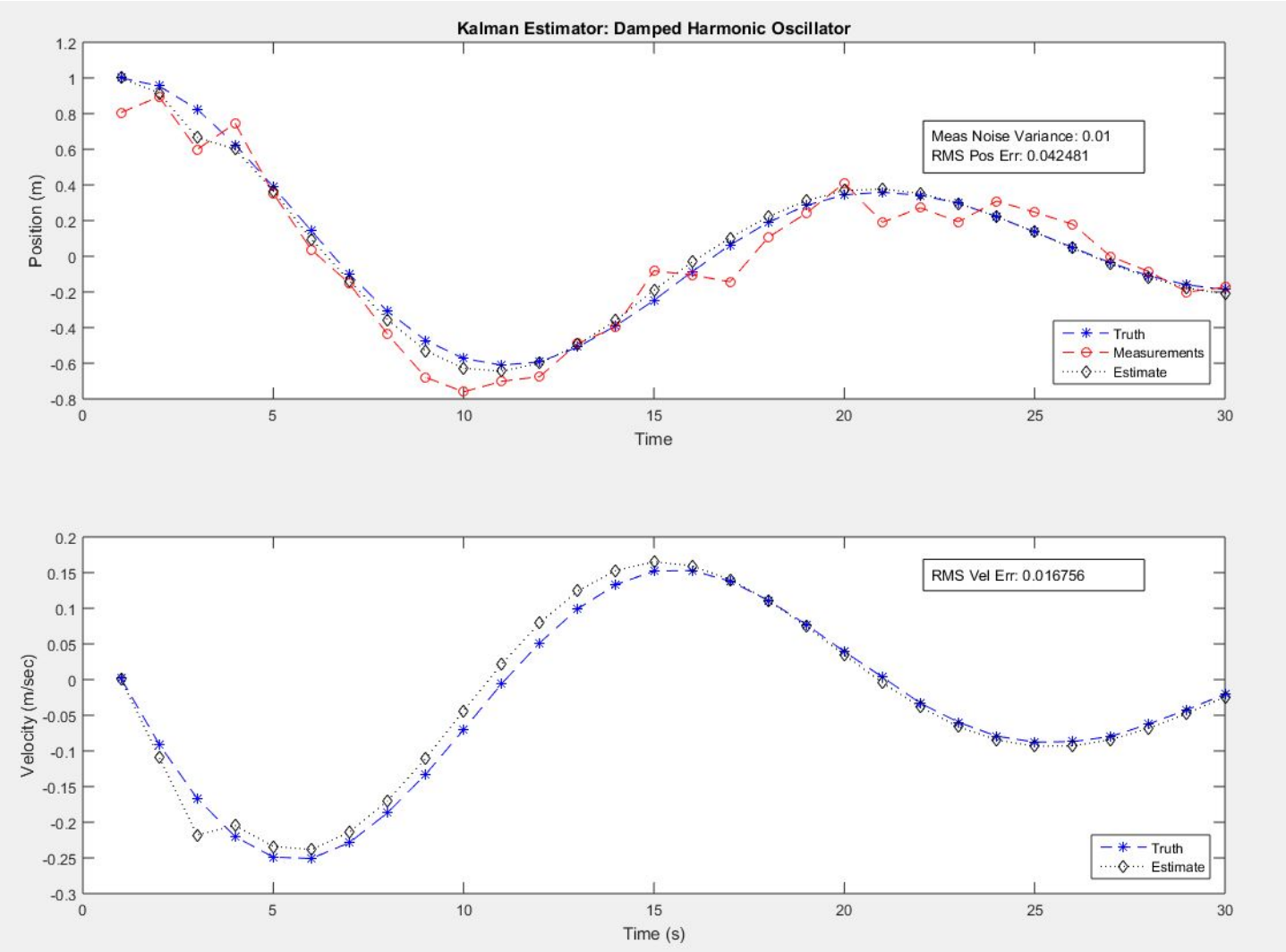
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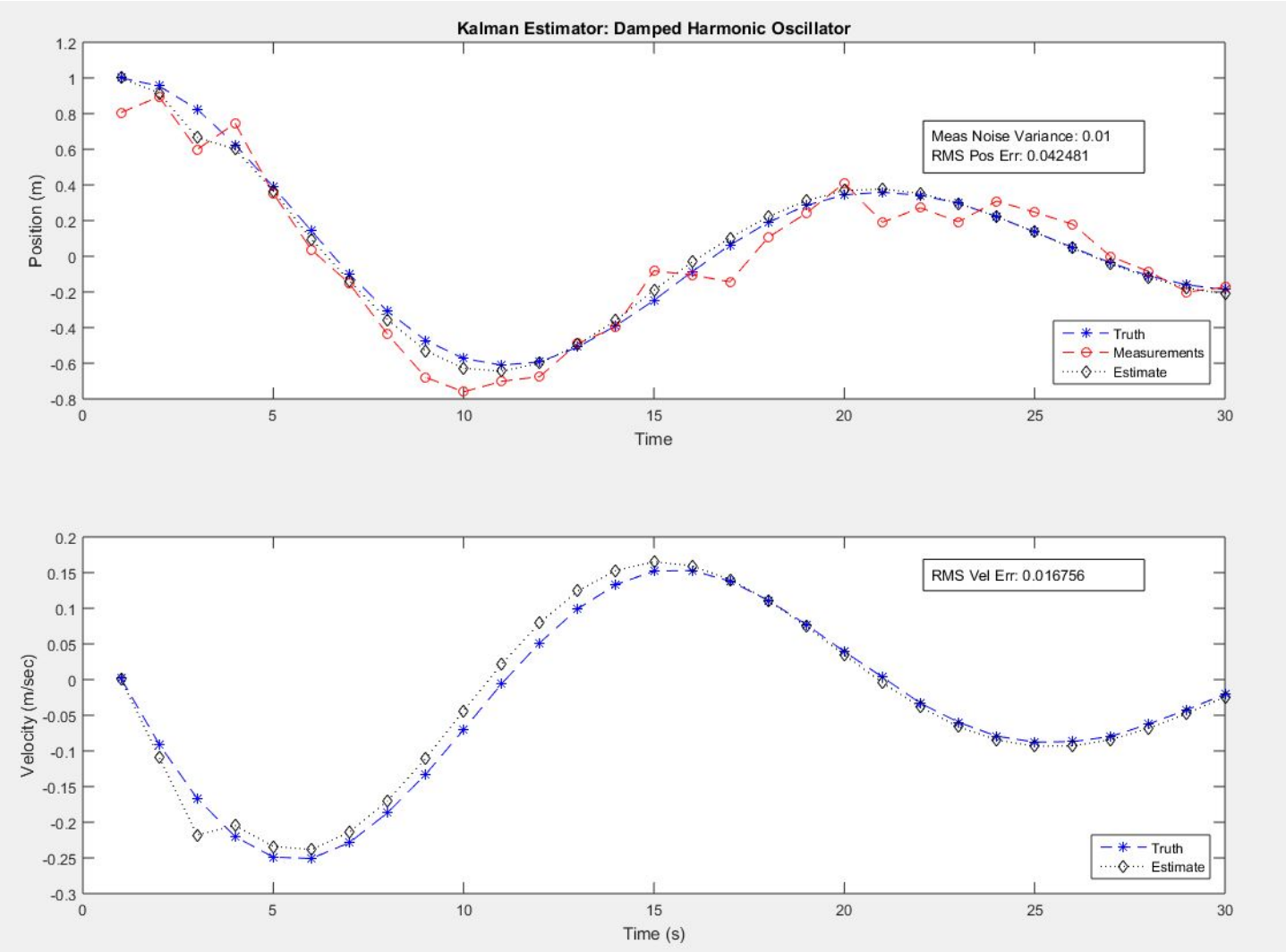
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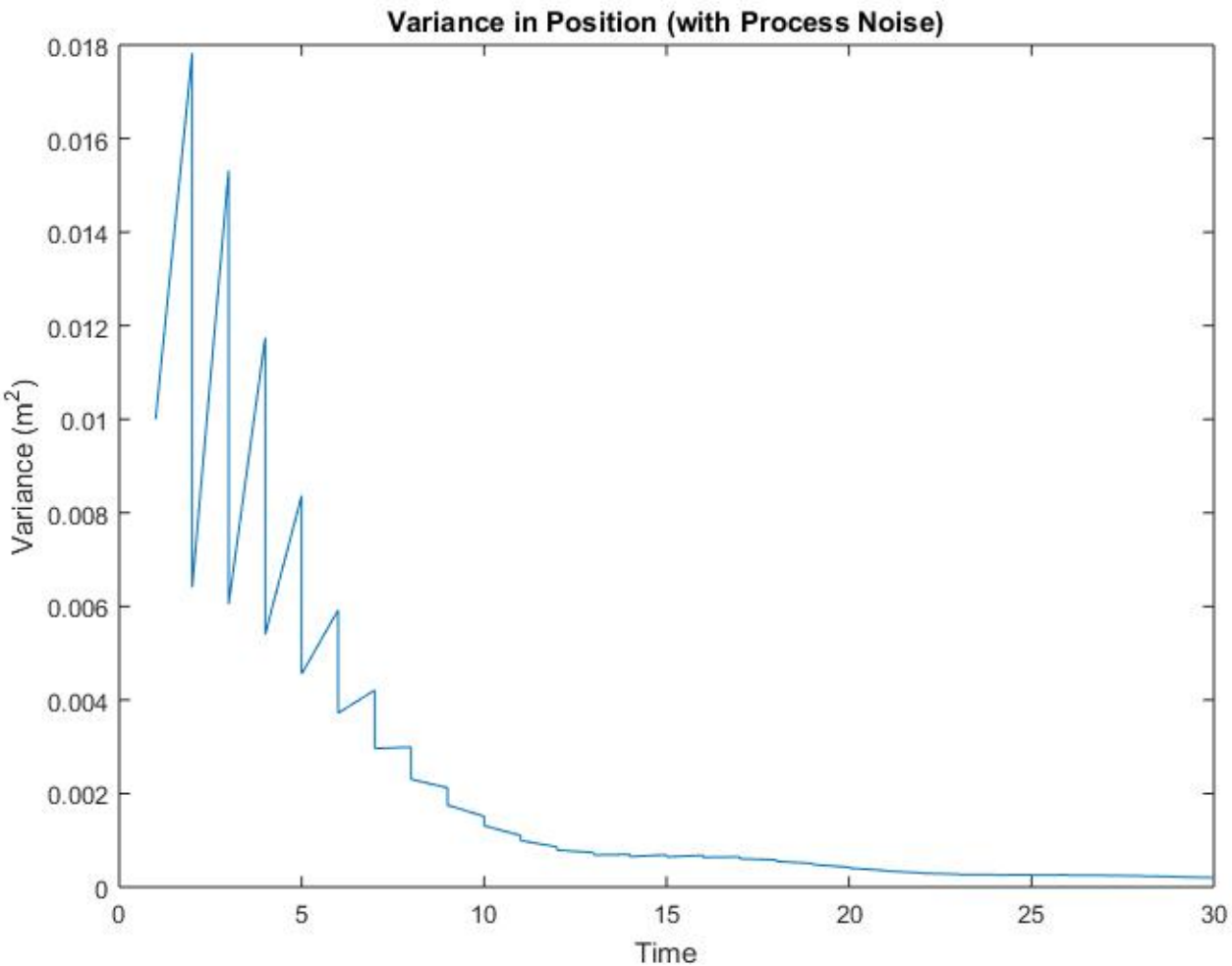
Damped Harmonic Oscillator Estimator: MATLAB Run



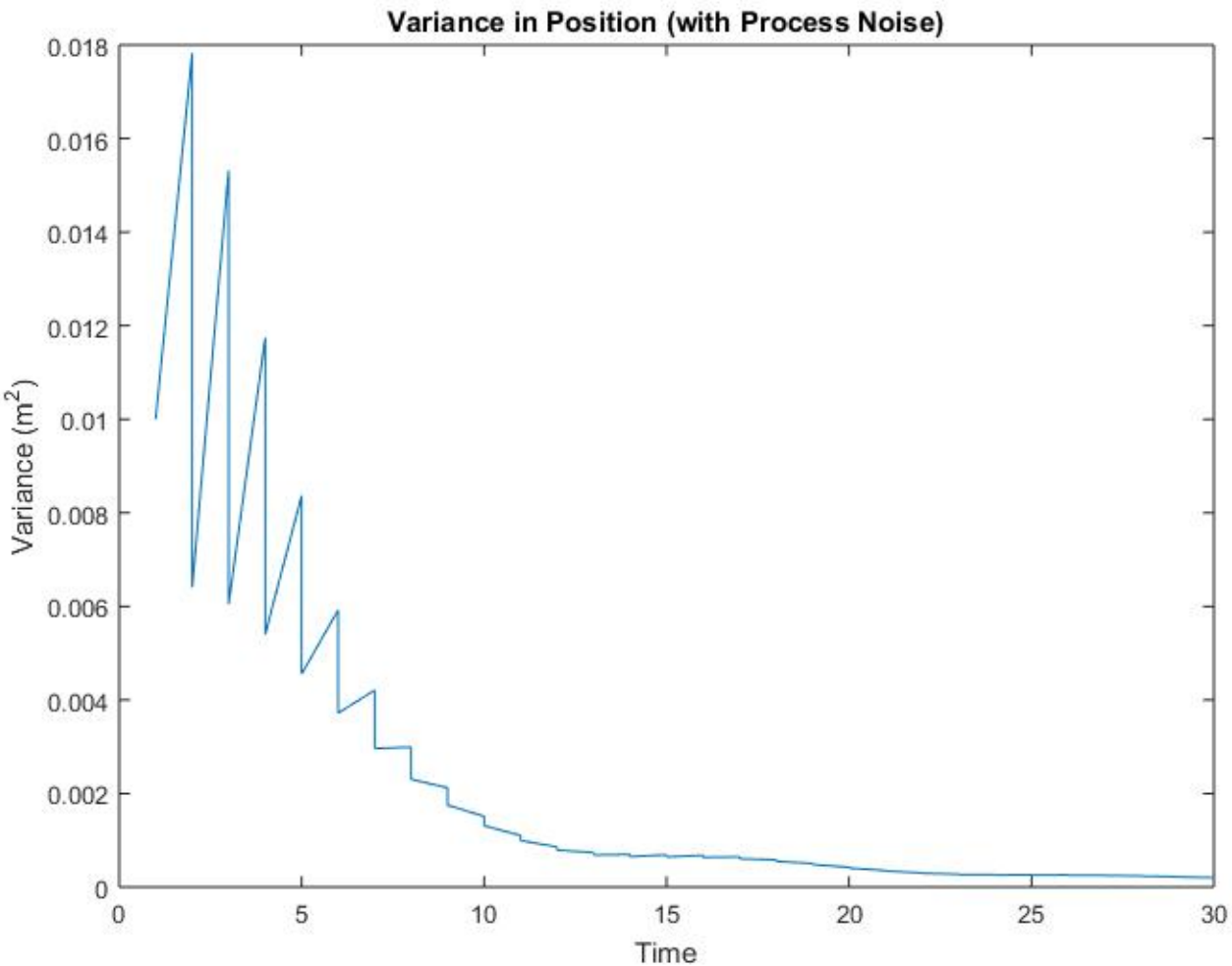
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Damped Harmonic Oscillator Estimator Sawtooth: MATLAB Run



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- Where are we?



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- Where are we?
  -
  -

- Where are we?
  -
  -
  -

- Where are we?
  -
  -
  -
- What's next?

- Where are we?
  -
  -
  -
- What's next?
  - TBD

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  -
  -
  -
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